Southern New Hampshire University

CS 255 Business Requirements Document Template

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of this project is for our consulting company to design and develop a fully managed online Driver's Education training and scheduling system for our client, DriverPass, to help achieve its goal of helping more people pass their Driving tests
- The Client, DriverPass, wants a system that allows student drivers to prepare for their driver's license test with different packages that provide online classes, practice tests, and on-the-road training.
- The system should allow students to schedule their reservations for driver training lessons by either completing the reservation online or through the office, and be able to track their progress with the online practice tests

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- The problem the owner of DriverPass noticed is that many people fail their driving tests at the DMV because of a lack of driver training.
- The company DriverPass proposes a solution to this by offering a way for these students to take online classes, practice tests, and on-the-road training in the form of "tiered packages" to help better prepare them for the DMV test.
- To achieve this vision, DriverPass needs a flexible, cloud-based web platform that hosts this online Student Driver's Education system to handle student registration, driver-student appointment reservations, online test/class progress tracking, and staying up to date with the latest updates from the DMV.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- Develop a web-based, cloud-hosted platform that works on computers and mobile devices

- Implement a scheduling system that allows either the student or secretary to reserve driving appointments for students with a tracking system that shows a specific driver, car, pick-up/drop-off location, and a 2-hour timeslot.
- Create a contact system that allows the student to contact the office, or the office to contact the student
- Design and create a student dashboard that tracks:
    - online test scores and statuses,
    - Driving instructor notes and comments
    - Student information and special needs
- Build an admin dashboard for the business owner and IT admin that allows them to:
    - manage accounts for employees, students, and drivers,
    - change user roles and permissions, block access, or reset passwords
    - easily disable package offerings that are no longer needed without a developer needed
- Build an exportable audit logging feature that tracks reservation activity, such as who made, modified, or cancelled the reservation
- Provide the ability for the owner to export/import reports and system data to be compatible with Excel for offline work
- Create and connect a notification system that connects with DMV to keep DriverPass up to date with the latest policies, rules, and sample questions.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system has to run in a cloud-based environment, so Liam and the students can access it from anywhere.
- The system needs to be able to handle multiple people booking driving sessions at the same time without lagging, crashing, or causing scheduling conflicts.
- The system should be updated regularly for security maintenance, but additionally, it should be updated whenever the DMV releases new rules, policies, or sample questions.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- This system needs to run on any standard operating system, but more importantly, be able to run on all modern browsers, since this is a web-based application.

- It needs to be mobile responsive, since there's a high probability that users will be logging in from their phones.
- The system will need a database, specifically a Relational database that will keep track of scheduling, user accounts, car assignments, test scores, driving notes, etc. Relational databases make it easier to join this data across several tables to be able to quickly look up the needed info.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system will distinguish users by creating accounts with specific roles (Admin, Student, Driver, Secretary) that would be tied to their login emails and user IDs. Both the user ID and email are unique and can distinguish users in the database.
- For logging in, the inputs won't be case sensitive, but passwords must be to preserve security and leverage hashing passwords for an added level of defense against hacking.
- The system will need to immediately inform the Admins or IT roles if there are any server errors, failed updates, or a hacking attempt. We can notify these roles using their emails, as well as in the Audit logs.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- The system will be designed to be able to disable packages by the Admin without needing to change code or call a developer.
- The IT admin needs full administrative access to add, remove, or modify employee and student accounts as needed. Additionally, it would be wise to give the IT admin access to system controls such as updates, logs, and server information in case any oddities arise after launching the app.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- The user will need a valid email and password to log in.
- The email will need to be verified once an account is created.
- Since this data exchange between client and server is over the web, we can leverage HTTPS and SSL web encryption to secure the data in transit.
- Any account that has failed to log in too many times will need to be locked out for at least 30 minutes. Sometimes users forget their passwords. So, the timer will help protect against hackers while also not aggressively locking out real users too often.
- If a user cannot remember their password, the IT admin can reset their password.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall allow students to schedule, cancel, and modify their 2-hour driving appointments online.
- The system shall track which user is assigned to which driver and car for specific timeslots.
- The system shall allow the owner of DriverPass to export data and reports into an Excel format for offline work.
- The system shall connect to the DMV to receive automated notifications about updated rules and sample questions.
- The system shall provide students with access to online classes, materials, and practice tests.
- The system shall keep track of students' class progress and test scores and allow them to view them.
- The system shall allow drivers to add notes to students' profiles.
- The system shall allow the secretary to schedule appointments for students if they call in or visit the office.
- The system shall have different packages that allow students to purchase different levels of access.
- The system shall record a pickup and drop-off location for each driving lesson.
- The system shall track and log user activity, including creating, cancelling, or modifying reservations.
- The system shall allow the Admin to print activity reports.
- The system shall allow the students to contact the office, and allow the office to contact the students online via messaging.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The interface needs to run on all modern browsers and be mobile-friendly.
- The main users are Liam (Owner/Admin), Ian (IT Admin), the secretary, the drivers, and, of course, the students.
- Students will need a dashboard to track their online course progress, including test scores, test names, time taken, score, and status, as well as driving appointments, drivers, cars, pickup and drop-off locations.
- The Drivers will need a dashboard to see their appointments, pickup and drop-off locations, which car they will be using, and the student's profile, along with any driver's notes on the student profile.
- The secretary will need an input form for customer registration, such as name, address, and credit card information.
- The Admin and IT admin will have access to system controls such as the activity reports page, audit logs, and a user table for resetting or blocking accounts.

- Users will interact with the system through the web, meaning they can access it from a desktop or mobile device, and they will be able to click through pages they have access to based on their roles and permissions.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- First, one big assumption is that the DMV has an API that allows the system to connect to for automatic updates to policies, etc.

- I'm assuming that all users will have reliable internet access, smartphones, and modern computers.

- I'm assuming that our Admin and IT admin are technically savvy and understand how to use the admin tools and audit logs described.

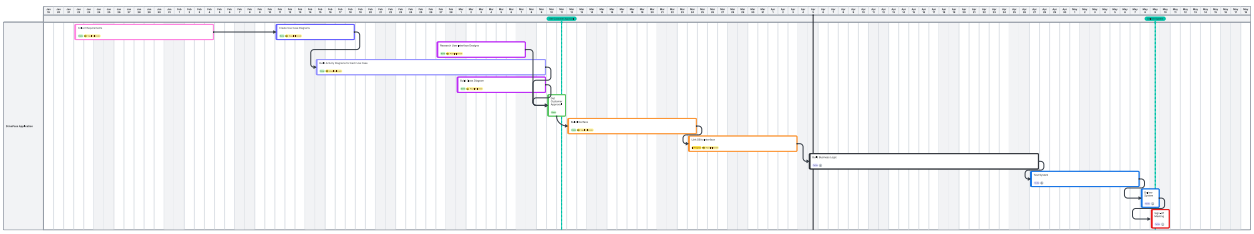
- Another assumption is that we won't need to make many changes after launching the system. Prices may change, curriculum, or even user interfaces might need to be adjusted after launch.

- Another assumption not addressed specifically in the design above is which cloud-based platform we are using; based on that decision, it can affect other decisions, costs, and technical implementations.

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- The biggest limitation is the project timeline; turning around a project like this in a couple of months is a big ask.
- The second major limitation is that the entire system is cloud-based, meaning any disruption to the internet would affect how the system works.
- The third limitation would be our resources in terms of developers and designers. If any member of the team gets sick or has to take time off, then the development of the system will be affected.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.